Candidate Name: _____

Class	Adm No

millennia institute

2026 Preliminary Exams

Pre-University 3

H2 PHYSICS **9749/02**

Paper 2 Structured Questions **31 August**

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not turn over this page until you are told to do so.

Write your full name, class and Adm number in the spaces at the top of this page.

Write in dark blue or black pen on both sides of this booklet.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.
Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question

.

For Examiner's Use		
1		/ 10
2		/ 9
3		/ 7
4		/ 9
5		/ 7
6		/ 8
7		/ 11
8		/19
Presentation		
Total		/ 80

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_o	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_o	=	8.85×10^{-12} F m^{-1}
		=	$(1/(36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	ρgh
gravitational potential	ϕ	=	$-\frac{Gm}{r}$
temperature	T/K	=	T/°C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}<c^2>¿$
mean translational kinetic energy of an ideal gas molecule	E	=	$\frac{3}{2}\ kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t$
	v	=	$\pm\omega\sqrt{x_0^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	1/R	=	$1/R_1 + 1/R_2 + \ldots$
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
radioactive decay	x	=	$x_0 \exp(-\lambda t)$
decay constant,	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** questions in the spaces provided.

1 A golf ball is hit from point A on the ground and moves through the air to point B. The path is illustrated in Fig. 1.1.

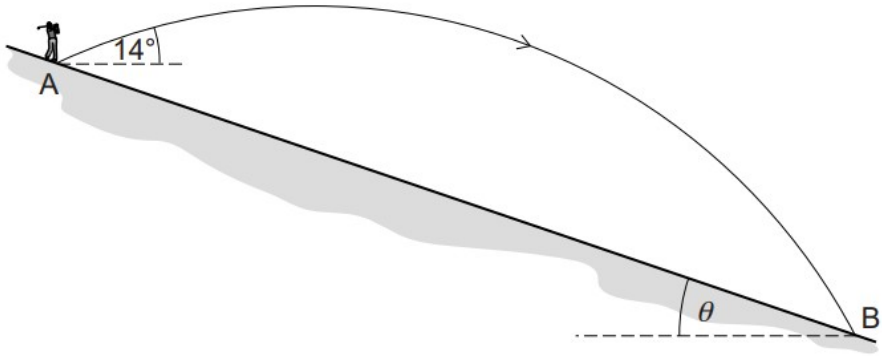


Fig. 1.1

The ground slopes downhill with a constant gradient. The ball has an initial velocity of 67 m s^{-1} at an angle of 14° to the horizontal. The ball hits the ground at B after 4.9 s.

(a) Ignoring air resistance, calculate

(i) the horizontal and vertical components of the ball's velocity at A,

horizontal component at A = ... m s^{-1}

vertical component at A = ... m s^{-1} [1]

(ii) the horizontal displacement from A to B,

horizontal displacement = ... m [2]

(iii) the vertical displacement from A to B,

vertical displacement = ……..... m [2]

(iv) and the angle of the slope to the horizontal, θ.

angle θ = ……..... ° [2]

(b) In reality, air resistance acts on the ball in the direction opposite to its motion.

(i) On Fig. 1.1, sketch a likely path of the ball hit from A when air resistance is taken into account.

[2]

(ii) Give reasons for the shape you have drawn in **(b)(i)** for the path of the ball at the start.

……..

……..[1]

[Total: 10]

2 A bungee jumper of mass 60 kg is attached to an elastic rope which starts to stretch after a short time of free fall. The gravitational potential energy of the bungee jumper is 0 J when she has fallen through 40 m to the lowest point.

(a)

	gravitational potential energy / kJ	elastic potential energy / kJ	kinetic energy / kJ
top		0	0
half-way		2.6	
bottom	0		

[3]

Fig. 2.1

Fill up the missing energies at the top, bottom and half-way positions in Fig 2.1. Drag forces can be considered negligible.

(b) Show that the unstretched length of the elastic rope is 10 m.

[2]

(c) Using **(b)**, determine the spring constant, *k* of the elastic rope.

k = ……N m^{-1} [2]

(d) Determine the extension of the elastic rope when the kinetic energy of the bungee jumper is the highest.

extension = …….. m [2]

[Total: 9]

3 Ice is less dense than liquid water due to the structure of the bonds between the molecules when it is in a solid state. Hence, when ice melts into water at 0°C, the density of water increases. As the liquid water increases in temperature, its density further increases to a maximum value at 4.0°C.

(a) Consider a 5.0 kg block of ice at 0°C that is melted into liquid water at the same temperature using a heat source supplying a constant power of 4180 W.

(i) The specific latent heat of fusion of ice is 3.34×10^5 J kg^{-1}.

Determine the time taken to completely melt the block of ice into liquid water at 0°C.

time = …….…….. s [2]

(ii) The density of ice at 0°C is 0.915 kg m^{-3} while the density of liquid water at 0°C is 0.999 kg m^{-3}.

Calculate the work done on the ice at a constant atmospheric pressure of 1.01×10^5 Pa as the ice melts completely into water at 0°C.

work done on the ice = ……..…… J [3]

(b) The first law of thermodynamics for a system can be expressed as

$$\Delta U = q + w$$

where ΔU is the increase in internal energy of the system, q is the heat supplied to the system and w is the work done on the system.

Use the words **positive**, **negative** and **zero** to complete Table 3.1 for the three terms in the equation for each of the processes shown. You may use each word once, more than once, or not at all.

Table 3.1

process	ΔU	q	w
ice melting into liquid water at 0°C			
liquid water warming up from 0°C to 4°C			

[2]

[Total: 7]

4 Two oppositely charged parallel metal plates P and Q are placed in vacuum. The electric field strength E is uniform in the region between the plates, and is zero outside this region. Uniform magnetic flux density B into the plane of the paper also exists in the region between the plates, as well as in the adjacent area, as shown in Fig. 4.1.

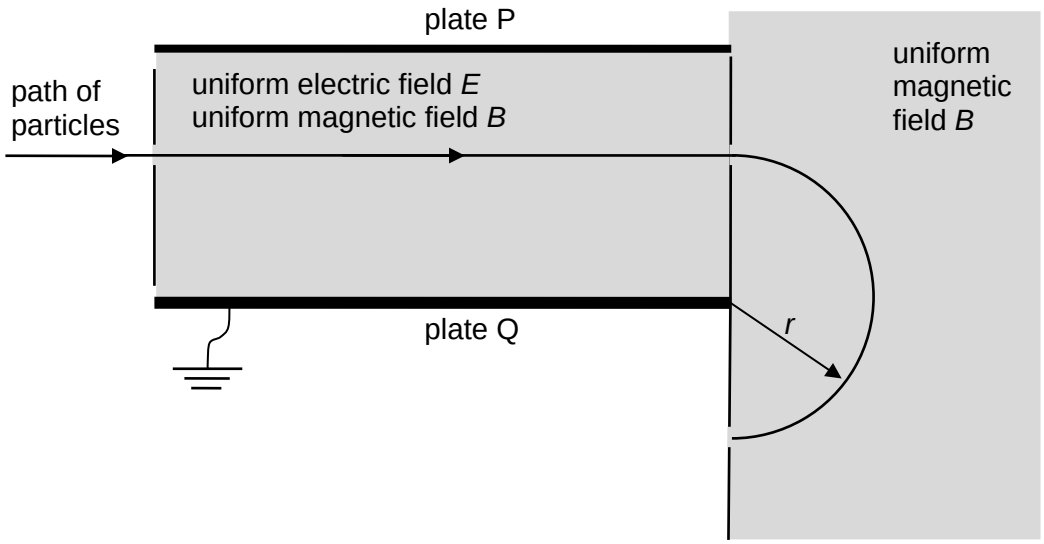


Fig. 4.1

A beam of negatively charged particles moving in the direction perpendicular to the electric field enters the region between the plates.

(a) **(i)** Some of the particles move in a straight line undeflected in the region between the plates.

State and explain the polarity, positive or negative, of plate P.

……

……

……

…….. [2]

(ii) Show that the speed v of these particles satisfies

$$v = \frac{E}{B}$$

[1]

(iii) $v > \frac{E}{B}$

Some of the particles move with an initial speed .

On Fig. 4.1, sketch the path between the plates of these particles.

Explain your answer.

……

……

…….. [2]

(b) After leaving the parallel plates, the undeflected particles enter a region with only the uniform magnetic flux density *B*, in which they follow a circular trajectory of radius *r,* as shown as Fig. 4.1.

(i) Explain why the speed of the particles remains constant along the circular trajectory.

……

……

…….. [2]

(ii) Show that, for the particles that follow the circular path of radius *r*, the charge-to-mass ratio is given by

$$\frac{q}{m} = \frac{E}{B^2 r}$$

.

[2]

[Total: 9]

5 An alternating supply of frequency 50 Hz and output 6.0 V r.m.s. is rectified to provide direct current for a resistor R, as shown in Fig. 5.1.

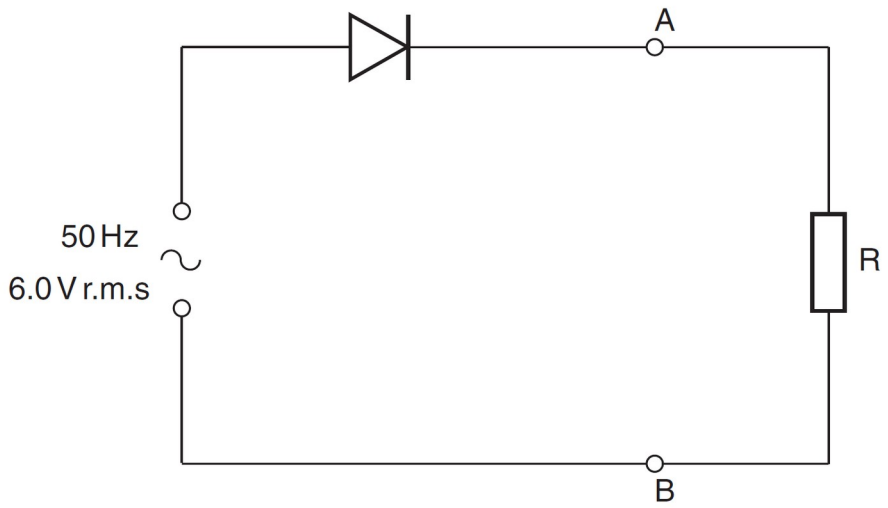


Fig. 5.1

The diode is ideal.

The output potential difference V of the alternating power supply is represented by

$$V = V_o \sin(\omega t)$$

where V_o is the maximum potential difference in volts, ω is the angular velocity in rad s^{-1} and t is the time in seconds.

(a) By reference to heating effect, state what is meant by the *root-mean-square (r.m.s.)* of an alternating current.

……

……
…

……
[1]

(b) Determine the value of V_o.

V_o = …… V [1]

(c) Determine the value of ω.

value of ω = ……rad s^{-1} [1]

(d) Determine the maximum potential difference across the diode during one cycle.

potential difference = …… V [1]

(e) State the potential difference across A and B when the diode has maximum potential difference across it. Give a reason for your answer.

……
…

……
[1]

(f) R has a resistance of 5.0 Ω .

In Fig. 5.2, sketch the variation with time of the power dissipated in R for **one** period of the alternating supply. Show the maximum power dissipated in R clearly in your graph.

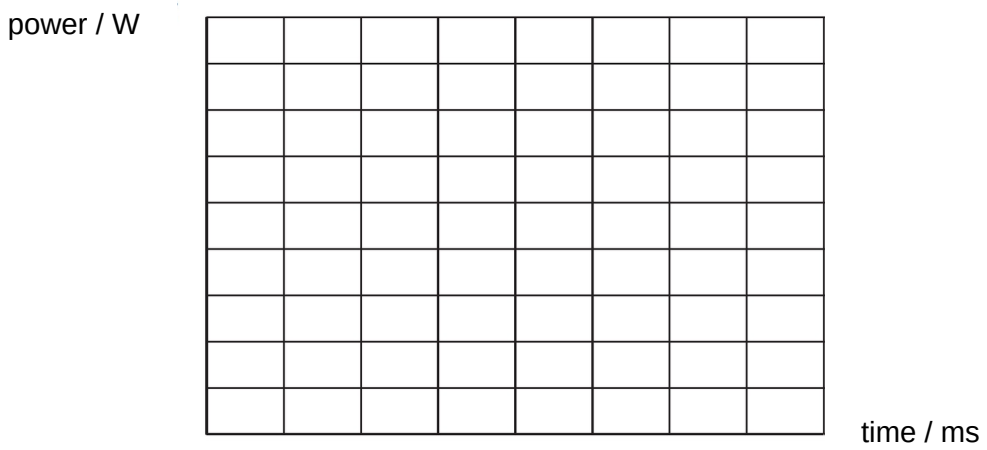


Fig. 5.2

[2]

[Total: 7]

6 (a) State how a line emission spectrum may be explained on the basis of the existence of discrete electron energy levels in atoms.

……

……

……

……

……

……

……[3]

(b) The lowest six discrete energy levels for a hydrogen atom are shown in Fig. 6.1. The diagram is drawn to scale and the ground state is at -13.6 eV.

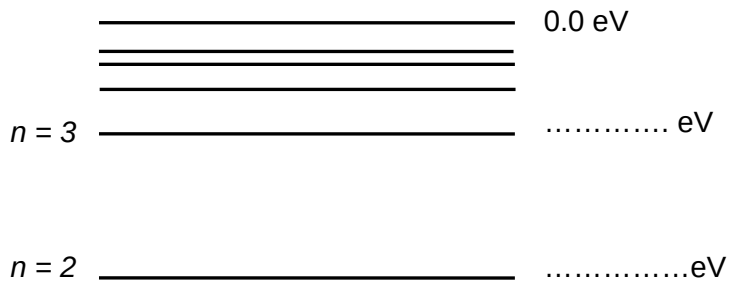


Fig. 6.1 (Drawn to scale)

(i) On Fig. 6.1, label the values for energy at energy levels $n = 2$ and $n = 3$. [2]

(ii) Calculate the wavelength of the light emitted when an electron falls from $n = 3$ energy level to the $n = 2$ energy level.

wavelength = …….. m [2]

(iii) Describe one method by which an electron in a gaseous hydrogen atom can be excited from a lower energy level to a higher energy level.

.....

.....[1]

[Total: 8]

7 **(a)** State what is meant by *radioactive decay*.

……
….

……
….

……
….

……..
[2]

(b) A radiation detector is placed close to a radioactive source. The detector does not surround the source.

Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different.

Suggest two other reasons why the activity and the measured count rate may be different.

1. ……..…
…

……..……
…

2. ……
…...

……..……..
[2]

(c) The variation with time *t* of the measured count rate in **(b)** is shown in Fig. 7.1.

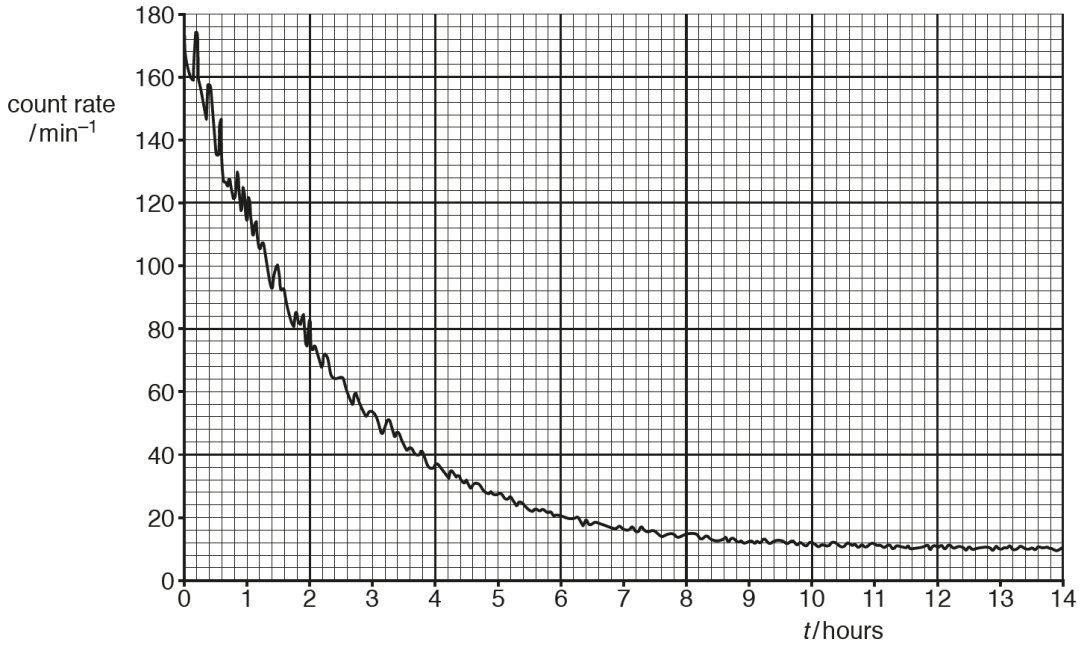


Fig. 7.1

(i) State the feature of Fig. 7.1 that indicates the random nature of radioactive decay.

……..……
…

……
[1]

(ii) Use Fig. 7.1 to determine

1. the background count rate and,

background radiation = …….. min^{-1} [1]

2. the half-life of the radioactive isotope in the source.

half-life = …… hours [3]

(d) The readings in **(c)** were obtained at room temperature.

A second sample of this isotope is heated to a temperature of 500 °C. The initial count rate at time $t = 0$ is the same as that in **(c)**. The variation with time t of the measured count rate from the heated source is determined.

State, with a reason, whether the heating will cause a difference, if any, in the half-time.

……
….

……
….

……
[2]

[Total: 11]

8 Read the passage below and answer the questions that follow.

X-ray and magnetic resonance imaging (MRI) are some modern imaging techniques in medicine that uses externally placed devices to obtain diagnostic information from underneath the skin.

A modern form of X-ray tube used to obtain the internal body structure of a patient is shown in Fig. 8.1 (not to scale).

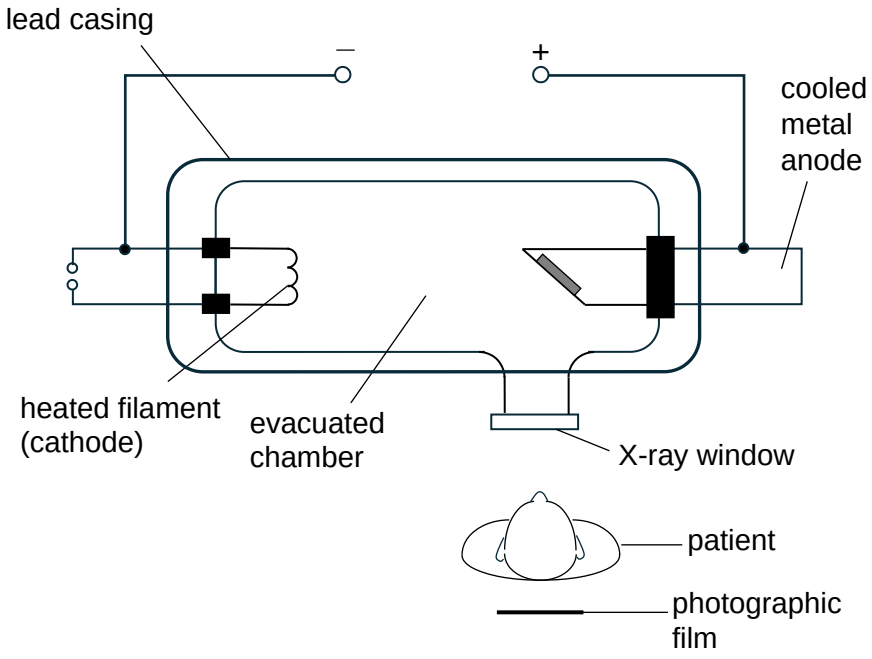


Fig. 8.1

In the X-ray tube, electrons emitted from the heated filament are accelerated through a large potential difference towards the metal anode, producing X-rays. The filament at the cathode is

typically made of thin tungsten wire. The X-rays produced are controlled and directed to leave the X-ray tube via a window. As the X-ray beams pass through the patient, varying degree of X-ray gets absorbed, depending on the composition of the body. The remaining X-rays then reach the photographic film where a contrast image of the internal body structure is obtained. Good contrast is achieved if there is a clear difference in the blackening of the photographic film as the X-ray passes through and gets absorbed by different types of tissue in the patient. Typically, a good contrast is obtained when the ratio of the transmitted X-rays between different body parts has an order of magnitude of at least 1.

The gradual decrease in intensity of a beam of X-ray as it passes through matter is represented by the equation

$$I = I_0 e^{-\mu x}$$

where I_0 is the initial intensity, x is the thickness of the material, I is the transmitted intensity and μ is the absorption (attenuation) coefficient. Fig. 8.2 shows the absorption (attenuation) coefficient of some matter with 30 keV X-rays.

matter	μ / cm^{-1}
blood	0.41
bone	2.46
brain	0.41
muscle	0.40

Fig. 8.2

Magnetic resonance imaging (MRI) is another imaging technique that relies on the fact that some atomic nuclei behave like tiny magnets in an external magnetic field.

In MRI, it is usually the nuclei of hydrogen atoms that are studied, since hydrogen atoms are present in all tissues. The hydrogen nucleus contains only one proton, and it has a property called spin. When a very strong external magnetic field is applied, the magnetic axis of a hydrogen nucleus does not align itself directly along the external magnetic field, but rotates around it, just like the axis of a spinning top, as shown in Fig. 8.3. This rotation or gyration action is known as precession.

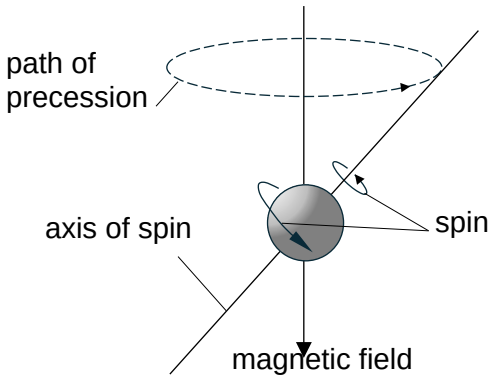


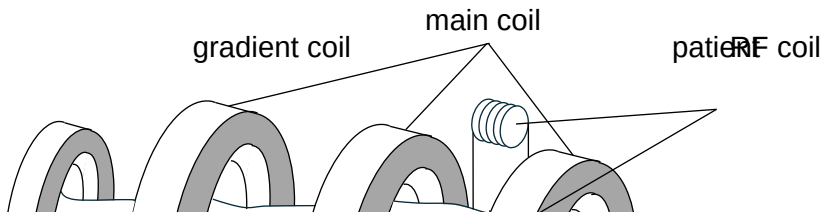
Fig. 8.3

The angular frequency of precession is called the larmor frequency, ω_0, and depends on the individual nucleus and the magnetic flux density B_0 of the magnetic field.

$$\omega_0 = \gamma B_0$$

The quantity *y* is the gyromagnetic ratio for the nucleus and is a measure of its magnetism. *y* for hydrogen nucleus is approximately 2.68×10^8 rad s^{-1} T^{-1}.

Fig. 8.4 shows an MRI scanner comprising three set of coils: main coil, gradient coil and radio frequency (RF) coil. The main coil is a solenoid that is 2.2 m long and 1.0 m in diameter. It is made of superconducting wire that carries a constant current of 750 A and produces an external magnetic field of 1.5 T.


main coil
gradient coil
patient
RF coil

(a) Explain the principles of production of the continuous X-ray spectrum.

……..

……..

……..

……..

……..

……..[3]

(b) Suggest a reason why tungsten is used for the filament at the cathode of an X-ray tube.

……..

……..[1]

(c) A cross-section of a model arm is shown in Fig. 8.5. In an investigation into the absorption of X-ray radiation in the model arm, parallel X-ray beams of 30 keV are directed along the line MM and along the line BB.

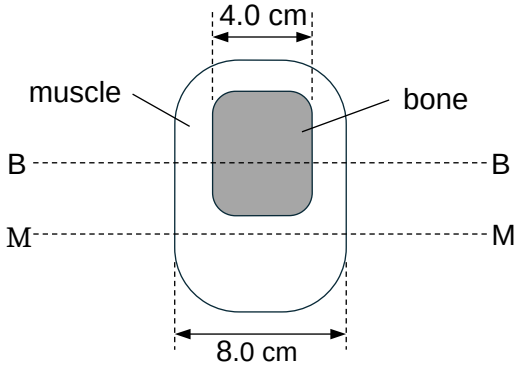


Fig. 8.5

(i) Calculate the ratio

$$\frac{\text{intensity of transmitted X-ray beam from model}}{\text{intensity of incident X-ray beam on model}}$$

for a parallel X-ray beam directed along the line

1. MM,

ratio = …… [2]

2. BB.

ratio = …… [2]

(ii) Explain whether the X-ray image obtained has good contrast.

……..

……..

……... [2]

(d) MRI scanning typically takes 30 minutes to more than 1 hour. State one other disadvantage of using MRI.

……..

……..[1]

(e) Determine the number of turns per unit length, *n* in the main coil of the MRI scanner.

n = ……..m^{-1} [2]

(f) Determine the larmor frequency, ω_0, of the pulse of the RF waves required to cause nuclear magnetic resonance of hydrogen nuclei in the MRI scanner.

ω_0 = ……..…….... rad s^{-1} [2]

(g) Using **(f)**, determine the frequency of the RF waves.

frequency = …….. Hz [2]

(h) During an MRI procedure, a small segment of the main coil loses its superconductivity, and the resistance of the wire suddenly increases to 0.0045 Ω. An increase in temperature of the main coil occurs, causing liquid helium to rapidly vaporise. The latent heat of vaporisation of helium is 21 kJ kg^{-1}.

Determine the initial rate of vaporisation of liquid helium.

initial rate of vaporisation = …….. kg s^{-1} [2]

[Total: 19]

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